LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	44
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief explanation of about typical ADI, HSI and flight computer. Brief explanation of EFIS and concept of symbology.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief explanation of about typical ADI, HSI and flight computer. Brief explanation of EFIS and concept of symbology.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of instrument symbology diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the Attitude Director Indicator?
(b) How does a Horizontal Situation Indicator differ from a standard compass?
(c) Define the term Electronic Flight Instrument System.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Typical ADI components and indications • HSI display elements and navigation • Flight computer calculation basics	L & Ds	PPT, IB	15 min
(b)	• EFIS architecture and display types • Primary flight symbology concepts • Secondary flight symbology concepts	L & Ds	PPT, PDF	20 min
(c)	• Integration of AFCS with displays • Symbology interpretation for flight control • System limitations and redundancy	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the Flight Computer assists the pilot in navigation calculations.
(b) Describe the symbology used on an EFIS Primary Flight Display.
(c) Differentiate between the functions of ADI and HSI in a typical aircraft.

12. Home Assignment and Follow Up	Prepare a brief note on the architecture of a typical FBW system to be discussed in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Architecture and control laws of a typical FBW system. Advantages and disadvantages of an FBW system.	Refer to the master syllabus reference list	LP 45

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min